

Lec 9: Random Variables.

Given a probability model $(\Omega, \mathcal{F}, P)$.
Random Variables are defined as a function from the sample space Ω to the set of real number.

$$X: \Omega \rightarrow \mathbb{R}.$$

a

$$\Omega = \{H, T\}^3 \quad (3 \text{ coin tosses})$$
$$= \{HHH, THH, \dots\}.$$

$$X(\omega) = \# \text{ heads in } \omega \quad \text{for all } \omega \in \Omega.$$

$$\text{For example } X(HHT) = 2.$$

The values
X can
take are
finite
of values

$$\Omega = [0, 1]$$

$$X(\omega) = \begin{cases} 0 & \omega \leq 0.5 \\ 1 & \omega > 0.5 \end{cases}$$

uncountable
many outcomes
for X

$$\Omega = [0, 1]$$

$$X(\omega) = \omega \quad \omega \in [0, 1]$$

d

$$\Omega = \{H, TH, TTH, \dots\}$$

1 2 3

$$X(\omega) = \text{length of } \omega.$$

$$X(TTH) = 3.$$

Countable
but not
finite

of values that a R.V can take can be
countable or uncountable.

We are interested in looking at the probability that this r.v. takes some values.

$$P(X=x) \triangleq P(\{\omega : X(\omega) = x\}).$$

For this probability to be well-defined we need $\{\omega : X(\omega) = x\} \in \mathcal{F}$.

Suppose we consider the example

$$\Omega = [0, 1] \quad P((a, b)) = b - a, \quad X(\omega) = \omega.$$

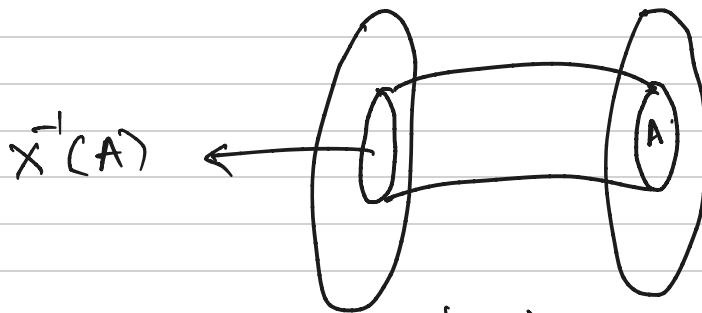
$$P(X=x) = P(\{x\}) = 0 \quad \forall x, \quad \text{So not meaningful to just look at } P(\{X=x\}).$$

In general, for a random variable to be well-defined over the probability model $(\Omega, \mathcal{F}, P)$ the following needs to hold true.

$$\{\omega : X(\omega) \leq x\} \in \mathcal{F} \text{ for all } x \in \mathbb{R}.$$

We will use the notation

$$X^{-1}((-\infty, x]) = \{\omega : X(\omega) \leq x\}.$$



In general $X^{-1}(A) = \{\omega : X(\omega) \in A\}.$

Lemma: If $X^{-1}((-\infty, x]) \in \mathcal{F}$ for all $x \in \mathbb{R}$.

then

① $X^{-1}((-\infty, x)) \in \mathcal{F}$

② $X^{-1}(\{x\}) \in \mathcal{F}$ $\rightarrow \{w \in \Omega \mid X(w) = x\}$

③ $X^{-1}([x_1, x_2]) \in \mathcal{F}$

④ $X^{-1}((x_1, x_2)) \in \mathcal{F}$.

Proof:

① By assumption $X^{-1}((-\infty, x]) \in \mathcal{F}$
for all $x \in \mathbb{R}$.

Let $A_n = (-\infty, x - \frac{1}{n}]$

$B_n = X^{-1}(A_n) \in \mathcal{F}$.

$\bigcup_{n=1}^{\infty} A_n = (-\infty, x)$

Clearly $\bigcup_{n=1}^{\infty} B_n \in \mathcal{F}$ (by closure under union property of $\mathcal{F}$)

$\bigcup_{n=1}^{\infty} B_n = \bigcup_{n=1}^{\infty} X^{-1}(A_n) \in \mathcal{F}$

$= X^{-1}\left(\bigcup_{n=1}^{\infty} A_n\right)$

$= X^{-1}((-\infty, x)) \in \mathcal{F}$.

②

$$X^{-1}(x) \in \mathcal{F}$$

$$X^{-1}(\underbrace{(-\infty, x]}_A) \in \mathcal{F} \text{ and from}$$

$$\textcircled{1} \quad X^{-1}(\underbrace{(-\infty, x)}_B) \in \mathcal{F}.$$

$$A \setminus B = A \cap B^c = \{x\}.$$

(Closure under
complement &
intersection)

$$\begin{aligned} X^{-1}(A) \cap X^{-1}(B^c) &= X^{-1}(A \cap B^c) \in \mathcal{F} \\ &= X^{-1}(\{x\}) \end{aligned}$$

Properties ③ & ④ left as exercises.

Discrete Random Variable (R.V)

A R.V whose range is a countable set is referred to as discrete (R.V)

$$X: \Omega \rightarrow \mathbb{R}.$$

$\mathcal{X} = \{X(\omega) \mid \omega \in \Omega\}$ is a countable set
if X is discrete R.V.

Probability mass function (PMF).

$$\begin{aligned} P_X(x) &= P(X = x) \\ &= P(\{\omega : X(\omega) = x\}). \end{aligned}$$

an element of event space and
is well define (due to the property
2 of earlier lemma).

Properties of PMF

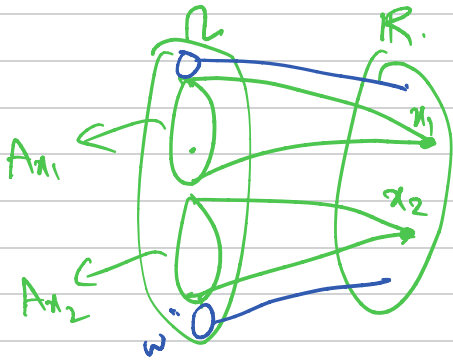
$$(a) \quad P_X(x) \geq 0 \quad \forall x \in \mathcal{X}_0$$

$$P_X(x) = P(X=x) = P(\{\omega: X(\omega)=x\}) \geq 0 \quad (\text{non-negativity axiom of probability})$$

$$(b) \quad \sum_{x \in \mathcal{X}_0} P_X(x) = 1.$$

$$\sum_{x \in \mathcal{X}_0} P_X(x) = \sum_{x \in \mathcal{X}_0} P(\underbrace{\{\omega: X(\omega)=x\}}_{:= A_x}).$$

$$A_{x_1} \cap A_{x_2} = \emptyset \text{ if } x_1 \neq x_2. \quad = P\left(\bigcup_{x \in \mathcal{X}_0} A_x\right) \quad (\text{additivity axiom of probability})$$
$$= P(\Omega) \quad (\text{R.V. takes values only in } \mathcal{X}_0).$$
$$= 1 \quad (\text{normalization}).$$



Examples.

$$(1) \quad \Omega = \{H, T\} \quad \mathcal{F} = \{\emptyset, \Omega, \{H\}, \{T\}\}.$$

$$P(\{H\}) = 1 - P(\{T\}) = p. \quad 0 \leq p \leq 1$$

$$X(\omega) = \begin{cases} 1 & \omega = H. \\ 0 & \omega = T. \end{cases}$$

$$P(X=1) = P(\{\omega=H\}) = p.$$

$$P(X=0) = P(\{\omega = \tau\}) = 1-p.$$

$$\textcircled{2} \quad \Omega = [0, 1] \quad P([a, b]) = b-a \quad 0 \leq a \leq b \leq 1$$

$\mathcal{F}$ = generated by open intervals (a, b) for $0 \leq a \leq b \leq 1$

$$X(\omega) = \begin{cases} 1 & \omega \leq p. \\ 0 & \omega > p. \end{cases}$$

$$P(X=1) = P(\{\omega : X(\omega) = 1\})$$

$$= P(\{\omega : \omega \leq p\})$$

$$= P([0, p]) = p.$$

$$P(X=0) = 1-p.$$

$X \sim \text{Bernoulli}(p)$ random variable has PMF

$$P(X=x) = \begin{cases} p & x=1 \\ 1-p & x=0. \end{cases}$$

Both the examples 1, 2 resulted in Bernoulli(p) random variable.

$\textcircled{3}$ Consider independent Bernoulli trials.

Say you are tossing a coin with bias p , n times.

All the tosses are independent of each.

$$\Omega = \{H, T\}^n \quad |\Omega| = 2^n.$$

A_i : event that i -th toss is head.

$$P(A_i) = p. \quad [\text{bias } p].$$

$A_1, A_2, \dots, A_n$ are independent of each other.

Binomial R.V.

$$X(\omega) = \# \text{ of heads in } \omega \quad \forall \omega \in \{H, T\}^n.$$

What are the values that the R.V. X can take?

$$\mathcal{X} = \{0, 1, 2, \dots, n\}$$

For example if $n=3$: $\Omega = \{HHH, HHT, HTH, \dots, TTT\}$
 $\leftarrow 8 \text{ elements} \rightarrow$

$$X(\omega) = \begin{cases} 3 & \omega = HHH. \\ 0 & \omega = TTT \\ 2 & \omega \in \{HTH, HHT, THH\} \\ 1 & \omega \in \{HTT, THT, TTH\} \end{cases}$$

$$\begin{aligned} P(X=0) &= P(\{\omega = TTT\}) = P(A_1^c \cap A_2^c \cap A_3^c) \\ &= P(A_1^c) P(A_2^c) P(A_3^c) = (1-p)^3. \end{aligned}$$

$$P(X=3) = P(A_1 \cap A_2 \cap A_3) = p^3.$$

$$P(X=1) = P(\{HTT, THT, TTH\}).$$

$$= P(\{HTT\}) + P(\{THT\}) + P(\{TTH\})$$

$$\begin{aligned} P(\{HTT\}) &= P(A_1 \cap A_2^c \cap A_3^c) \\ &= p(1-p)^2. \end{aligned}$$

$$P(\{THT\}) = P(1-p)^2 = P(\{TTH\}).$$

$$\Rightarrow P(X=1) = 3 \times p^1 \times (1-p)^2$$

$$P(X=2) = 3 \times p^2 \times (1-p).$$

$X \sim \text{Binomial}(n, p)$ random variable.

$$\mathcal{X}_0 = \{0, 1, \dots, n\}$$

For $k \in \mathcal{X}_0$

$$P(X=k) = \binom{n}{k} \times p^k \times (1-p)^{n-k}.$$

as

$$P(\underbrace{\{HH\dots H\}}_k \underbrace{\{TT\dots T\}}_{n-k}) = \underbrace{p \times p \times \dots \times p}_k \times \underbrace{(1-p) \times \dots \times (1-p)}_{n-k}$$

$$= p^k (1-p)^{n-k}$$

and there are exactly $\binom{n}{k}$ outcomes that have exactly k heads and all these outcomes have same probability.

$$\Rightarrow P(X=k) = \binom{n}{k} p^k (1-p)^{n-k}$$

Do these PMFs satisfy $P_X(x) \geq 0$

and $\sum_{x \in \mathcal{X}_0} P_X(x) = 1$.

① Bernoulli(p)

$$P_X(0) = 1-p, \quad P_X(1) = p. \quad \checkmark$$

$$P_X(0) + P_X(1) = 1.$$

Q Binomial (n, p)

$$P_X(k) = \binom{n}{k} p^k (1-p)^{n-k} \geq 0$$

for $k=0, 1, 2, \dots, n-1$.

$$\sum_{k=0}^n P_X(k) = \sum_{k=0}^n \binom{n}{k} p^k (1-p)^{n-k}$$

Using $(x+y)^n = \sum_{k=0}^n \binom{n}{k} x^k y^{n-k}$

$$(x+y)(x+y) \dots (x+y)$$

$$\sum_{k=0}^n P_X(k) = (p + (1-p))^n = 1$$